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D. What's the Diameter?

time limit per test: 1 second

memory limit per test: 1024 megabytes

You are given an **undirected connected** graph with N nodes and $N - 1$ edges. Your task is to find two nodes such that the path between those two nodes is the longest possible in the graph.

Input

The first line contains one integer N ($2 \leq N \leq 200000$) — the number of nodes.

The next $N - 1$ lines will contain two integers u_i, v_i ($1 \leq u_i, v_i \leq N$) — denoting there is a bidirectional road between u_i and v_i .

Output

On the first line, print a single integer — the length of the longest path.

On the second line, print two integers A and B — the nodes that form this longest path. If multiple pairs exist, you may print any one.

Examples

input	Copy
5 5 1 1 4 4 2 3 2	
output	Copy
4 3 5	
input	Copy
5 1 2 5 3 3 2 2 4	
output	Copy
3 5 1	
input	Copy
8 1 7 7 3 3 6 6 5 5 2 2 8 8 4	

CSE221 Assignment 06 Summer 2026

Contest is running

3 days

Contestant



→ Submit?

 Language: Java 21 64bit

Choose file:

[Choose File](#)

No file chosen



Success!


[Submit](#)

output

Copy

```
7
4 1
```

input

Copy

```
7
7 5
5 6
6 1
1 3
3 4
4 2
```

output

Copy

```
6
7 2
```

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The only programming contests Web 2.0 platform
Server time: Aug/20/2026 17:14:13^{UTC+6} (i1).
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